Systems Foundations

Deep Dive into DeepSeek — Chapter 19

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Foundation Books Project

The one rule of this chapter

A model can compute the right next-token distribution
and a serving system can still mishandle the request.

- The request may wait behind incompatible work, lose the cache state its next position needs, write into a page owned by a different sequence, or expose output before the producing device work has finished.
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Goal: build one reusable vocabulary — identity, ownership, completion, evidence — and follow it through CUDA execution, fabric movement, distributed ownership, and pipeline scheduling.

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5. What carries forward

One request, several simultaneous views

The live request systems record

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At systems step t, a live request systems record for request r is

$$\mathcal{S}_t(r) = \big(I_r, Q_t(r), W_t(r), C_t(r), O_t(r)\big).$$

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A component may transform one field, but must keep the result attached to the same request identity I_r.

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Issue is not completion: a producer's issued work becomes safe input only after its matching completion.

Execution coordinates, correctness, and V3 co-design

Same shape, different executable

$$\mathcal{V} = (r, c, p, e)$$

- r: repository. c: exact commit. p: built package or container. e: hardware/software environment that selected the executed path.
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$\mathcal{V}$ identifies the executable environment. The measurement protocol separately fixes workload, oracle, and timing window.

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Producer issued successfully	consumer read before matching completion

A visible token or correctly shaped tensor can pass while the hidden state the next component needs is already wrong.

Pressure transfer, not a free lunch

Definition (Co-design ledger)
$\mathcal{L}_d = (d,\ o_{\text{before}} \to o_{\text{after}},\ p_{\downarrow}, p_{\uparrow}, q, \text{e})$: the object that changed, the pressure relieved, the pressure added, the new obligation, and the evidence class.

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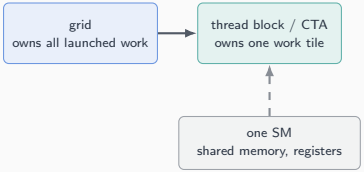
[REPORTED: DeepSeek-V3 technical report]: relieved pressure and added obligation are reported together — never one without the other.

CUDA and Tensor Core execution

From grid to lane: who owns what



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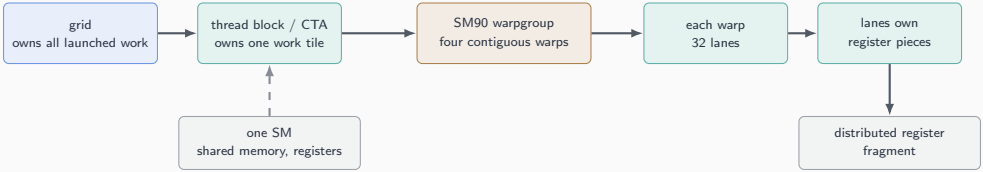

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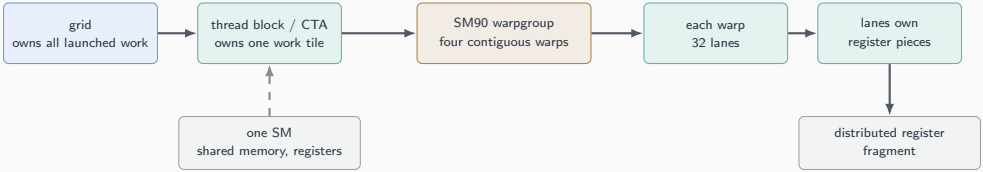

A CTA owns a work tile; a warpgroup names participating threads; a fragment names distributed per-lane state — three different nouns for three different things.

From concept to code: WGMMA's three operation-group boundaries

from DeepGEMM/deep_gemm/include/deep_gemm/ptx/wgmma.cuh

```
CUTLASS_DEVICE void warpgroup_arrive() {
    asm volatile("wgmma.fence.sync.aligned;\n" ::: "memory");
}
CUTLASS_DEVICE void warpgroup_commit_batch() {
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template <int N>
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Fence orders register access; commit closes a batch into a tracked group; wait drains pending groups before their accumulator is read. None of the three is the matrix operation itself.

[INSPECTED: DeepGEMM ptx/wgmma.cuh, revision 1f6f3f378920]

Two asynchronous engines, two completion facts

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Copy completion and matrix completion are never the same fact. A reusable stage needs both, plus the next owner's handoff, before reuse.

Fabric, distributed ownership, and pipelines

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Fabric words specialize the same handoff: object, interpretation, owner, completion, scope — now with participants, eligible buffers, and a local-or-cross-node path.

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DualPipe (Volume II) specializes this rule to bidirectional work, explicit transfers, and deferred weight-gradient actions.

What carries forward

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Identity, ownership, completion, evidence — one vocabulary,
reused across CUDA, fabric, distributed systems, and pipelines.

Next: CUDA kernels, fabric movement, and pipeline scheduling each specialize this same handoff with their own concrete objects (Volume II).